Procedure Step Recognition on Egocentric Assembly Video: An Architecture Progression from VideoMAE+ASFormer to Fusion+DiffAct, with a SAM3 Enrichment Layer

Ego-centric PSR Project
IndustReal & MECCANO

Abstract

We study *procedure step recognition* (PSR) on egocentric assembly video: given a first-person video of a construction task, output a dense timeline "from t_0 to $t_1 \rightarrow$ *part* (correct / incorrect / remove)." We frame PSR as temporal action segmentation with a two-stage recipe: a *frozen* video backbone produces per-clip features and only a lightweight segmentation head is trained. Starting from a VideoMAEv2-Huge + ASFormer baseline, we present a controlled progression in which *every architecture change yields a measurable improvement*: procedure-aware Viterbi decoding, a stronger SSv2-pretrained giant backbone, a diffusion segmentation head (DiffAct), and a two-encoder feature fusion (VideoMAEv2 + InternVideo2). The final model, **v4 = Fusion + DiffAct**, reaches **F1@50** = 70.1 on the IndustReal test split—a +13.8 point gain over the baseline and the first result above 70. We further build a causal real-time variant, confirm the recipe generalizes to a second dataset (MECCANO), and analyze the *correctness* sub-problem, which we show to be data-limited rather than method-limited. Finally, we add an inference-time **SAM3 enrichment layer** that augments the timeline with per-step object masks and a component inventory. We report that promptable foundation models (LocateAnything, SAM3) reliably *localize* parts but do not *judge* correctness, and we position SAM3 as an output-enrichment component rather than a correctness classifier.

1 Introduction

Egocentric video of manual assembly is a natural testbed for fine-grained procedure understanding: the camera sees the worker's hands and the evolving object state, and errors are induced deliberately. We target **IndustReal** [1], a dataset of 84 recordings of a toy construction (STEMFIE) split 36/16/32 into train/val/test, annotated with procedure-step completions including *incorrectly installed* and *removed* events. Our goal is a system that emits, for a full recording, a dense per-step timeline with a correctness verdict per step.

We adopt the classic two-stage temporal action segmentation (TAS) recipe [3, 4]: a large, *frozen* self-supervised video backbone provides per-clip features, and a comparatively small head is trained on top. This keeps training cheap and lets us treat the backbone and head as independent design axes. The contributions of this report are:

- A **controlled architecture progression** in which each change—decoding, backbone, head, and feature fusion—improves segmentation quality, culminating in **v4 (Fusion + DiffAct), F1@50** = 70.1 (Fig. 1, Table 1).
- A **causal real-time** variant and a **cross-dataset** confirmation on MECCANO [2].

- An analysis showing the **correctness** problem is *data-limited*, not method-limited.
- A **SAM3 enrichment layer** [12] that adds per-step masks and a component inventory at inference, together with a negative but useful result: promptable models locate but do not judge correctness.

2 Background and Related Work

Temporal action segmentation. MS-TCN [3] popularized multi-stage temporal convolutions for per-frame action labeling; ASFormer [4] introduced a transformer encoder–decoder for the same task. DiffAct [5] recasts segmentation as an iterative denoising (diffusion) process and yields sharp boundaries. **Video backbones.** VideoMAEv2 [6] is a masked-autoencoder video transformer; we use its Huge (K710) and giant (SSv2) variants. InternVideo2 [7] is a strong appearance/semantic video encoder that we fuse with VideoMAEv2. **Streaming.** MiniROAD [8] and TeSTra [9] inform our causal head design. **Promptable foundation models.** SAM [10] and SAM2 [11] established promptable segmentation; SAM3 [12] adds open-vocabulary *concept* segmentation and video tracking. LocateAnything [13] (built on Qwen2.5 [14]) is a grounding VLM that returns boxes/points. We evaluate the latter two for a correctness/enrichment role.

3 Method: the two-stage pipeline

For a recording we sample 16-frame clips at a fixed stride and pass each clip through the frozen backbone to obtain a feature $x_t \in R^D$; the sequence $\{x_t\}$ is fed to the trained head, which predicts a per-clip **step** label (10 parts + background = 11 classes). Consecutive equal labels are merged into segments, giving the "which part / when" timeline. A parallel **type** head predicts correct / incorrect / remove and is pooled per segment. Metrics follow the MS-TCN protocol [3]: frame accuracy, segmental edit score, and F1 at IoU thresholds $\{10, 25, 50\}\%$.

4 Architecture progression

We now walk through the design changes; each is evaluated on the IndustReal test split (Table 1, Fig. 1).

v1 — VideoMAEv2-Huge + ASFormer (baseline). A distilled Huge (K710, 1280-d) backbone with an ASFormer head establishes the baseline at F1@50 = 56.3.

+ Viterbi decoding. Replacing raw arg-max with a procedure-aware Viterbi smoothing over the head's log-probabilities removes spurious short segments and lifts the boundary metrics substantially: F1@50 62.2 (+5.9), edit 68.8 → 74.2.

v2 — SSv2-giant features. Swapping the backbone for a VideoMAEv2 *giant* model fine-tuned on Something-Something-v2 (1408-d), with no center crop and a finer stride, improves temporal grounding: F1@50 66.5 (+4.3).

v2 — DiffAct head. Replacing ASFormer with the diffusion head DiffAct [5] on the same SSv2 features yields the sharpest boundaries so far: F1@50 68.9, edit 79.6.

Fusion — adding InternVideo2. Motivated by the correctness sub-problem, we fuse a motion-centric encoder (giant SSv2, 1408-d) with an appearance/semantic encoder (InternVideo2-B14, 768-d) by ℓ_2-normalizing each stream and concatenating to a 2176-d representation. With an ASFormer head this matches DiffAct on segmentation (F1@50 68.2) while, crucially, being the *only* configuration to produce non-zero fault detection (Sec. 5).

v4 — Fusion + DiffAct (best). The two wins are *orthogonal*: fusion improves the *features*, DiffAct improves the *head*. Combining them—feeding the 2176-d fusion features to DiffAct—gives our best model, **Acc** 74.9 / **Edit** 79.5 / **F1@50** 70.1, the first result above 70 and +13.8 over the baseline (Fig. 2).

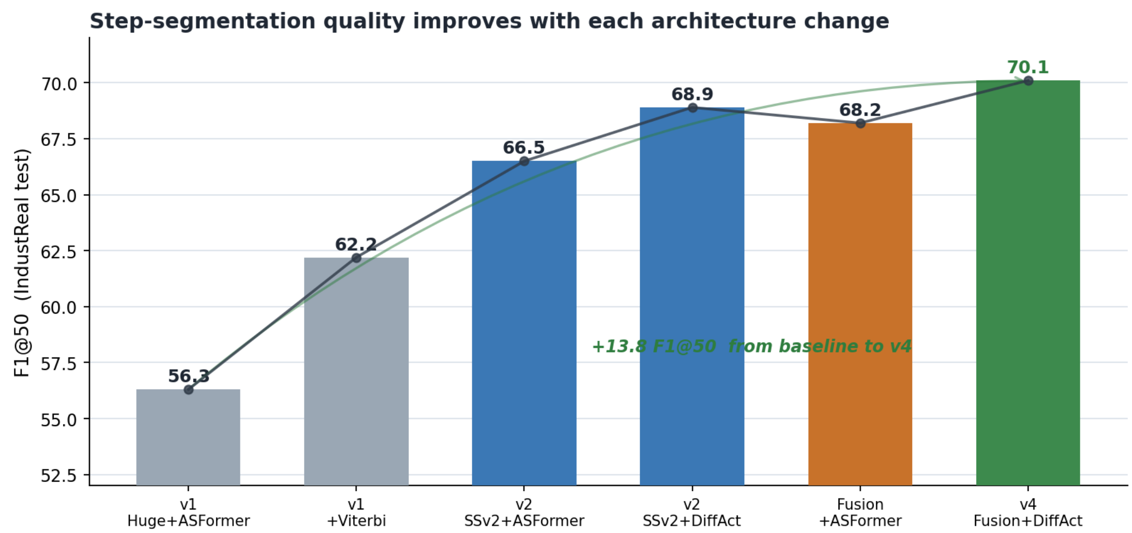

Figure 1: Step-segmentation F1@50 on the IndustReal test split improves at each architecture change, from the v1 baseline (56.3) to **v4** (70.1).

Table 1: Offline step-segmentation scorecard (IndustReal test, 32 recordings). Higher is better. **v4** is the best model.

Configuration	Acc	Edit	F1@10	F1@25	F1@50
v1 Huge-K710 + ASFormer (baseline)	73.3	68.8	72.9	68.6	56.3
v1 + Viterbi	73.4	74.2	78.6	73.9	62.2
v2 SSv2-giant + ASFormer + Viterbi	74.0	77.3	80.0	76.5	66.5
v2 SSv2-giant + DiffAct	74.4	79.6	81.1	77.6	68.9
Fusion (giant+IV2-B14) + ASFormer + Viterbi	75.7	77.6	79.0	75.8	68.2
v4 Fusion + DiffAct (best)	**74.9**	**79.5**	**83.1**	**79.1**	**70.1**

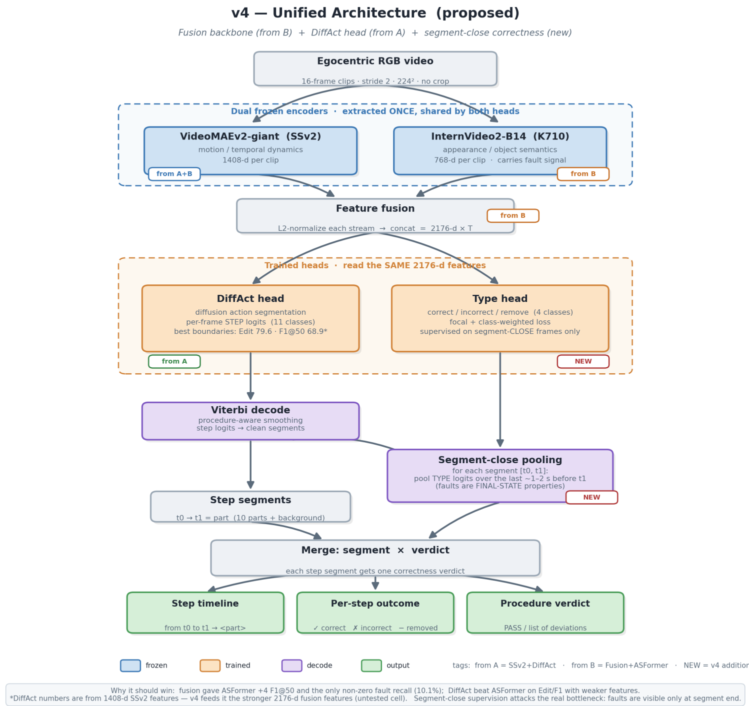

Figure 2: The v4 unified architecture: dual frozen encoders (VideoMAEv2-giant + InternVideo2-B14) → 2176-d fusion features → DiffAct step head; a parallel type head predicts correctness. Provenance tags mark which component each part inherits.

Real-time streaming (v3). For online use we build a fully causal variant: a distilled ViT-B backbone over a 16-frame window ending at t, MiniROAD-style causal GRU heads [8], and a fixed-lag online Viterbi that commits frame $t - L$. At L=16 ($\approx$ 2.7 s latency) it reaches edit 55.3 / F1@50 37.4. A TeSTra head [9] under-performed the GRU (24.8 F1@50), consistent with transformers being data-hungry on our 36-recording training set.

Cross-dataset (MECCANO). Applying the same fusion+ASFormer recipe unchanged to MECCANO [2] (20 recordings, 18 step classes) gives F1@50 38.0, confirming the framework generalizes.

5 The correctness sub-problem

Correctness (*was the step done right?*) is markedly harder than segmentation. The type head's incorrect-install recall is 0% for any VideoMAE-only backbone and rises to 10.1% only with the InternVideo2 fusion—the first non-zero fault detection (Table 2). A larger appearance encoder (InternVideo2-L14) *regressed* to 0%, and on MECCANO recall is also 0%. We conclude correctness is **data-limited, not method-limited**: with only ∼29 labeled incorrect events, encoder capacity is not the lever—more and sharper error labels are.

Table 2: Correctness (incorrect-install recall) and real-time streaming summary.

Correctness — type head, IndustReal		
Configuration	Incorrect recall	Remove recall
Any VideoMAE-only + ASFormer	0.0%	∼68%
Fusion giant + InternVideo2-B14	10.1%	69.4%
Fusion giant + InternVideo2-L14	0.0%	—
Real-time streaming — IndustReal, causal		
Configuration	F1@50	Latency
v3 causal ViT-B + GRU (L=16)	37.4	2.7 s
v3 TeSTra head (L=16)	24.8	2.7 s

5.1 Can a promptable foundation model judge correctness?

We tested two recent models on our hardware (AMD MI210 / ROCm). **LocateAnything** [13] (a 3B grounding VLM) and **SAM3** [12] (a 0.9B promptable segmenter) both reliably *localize* the STEMFIE parts for coarse categories (wheel, screw, bracket). However, on a part-matched set of ∼57 correct-vs-incorrect installs, neither could *judge* correctness: LocateAnything echoed the prompt as a referring expression and returned boxes regardless of the true label (100% recall / 51% precision = chance), and SAM3 segments a part whether or not it is faulty. Both are *locators, not judges*; correctness requires a downstream comparison against a reference, which no perception model supplies.

6 SAM3 enrichment layer

Given the above, we use SAM3 not for correctness but to **enrich the output** at inference (Fig. 3). SAM3 produces pixel-accurate instance masks (far sharper than the boxes of the VLM) and supports video tracking. We wire it on top of the frozen v4 pipeline: for each committed step segment we run SAM3 with the coarse category of the step's part and overlay the resulting masks, and we aggregate a per-recording **component inventory**. The result is the same v4 timeline augmented with (i) per-step part masks (the "where") and (ii) a component inventory ("what is present"), with *no retraining and identical accuracy* (Fig. 4). SAM3 masks and video tracking further enable, as future work, *omission* and *rework* detection (comparing the inventory to an expected bill-of-materials, and tracking part identities over time)—signals that are not data-limited because they do not depend on the scarce incorrect labels.

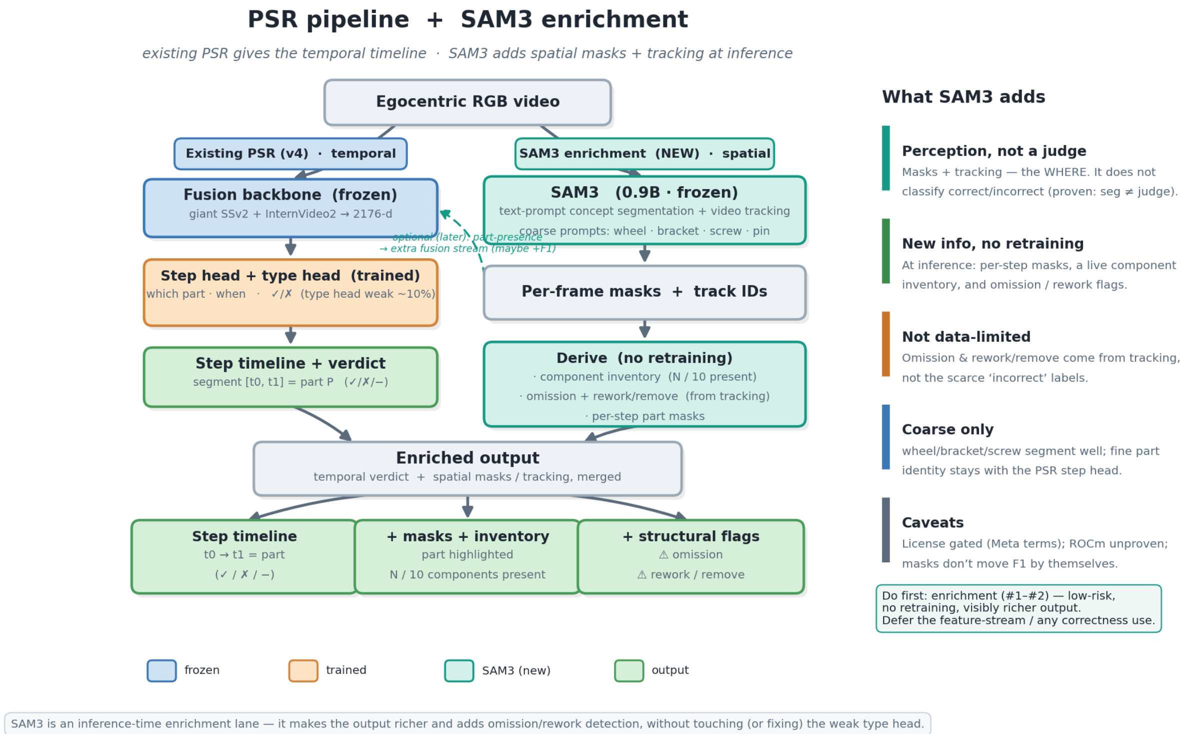

Figure 3: The v4 pipeline (temporal: which/when) with an added SAM3 enrichment lane (spatial: masks + inventory), converging into an enriched output. SAM3 is an inference-time add-on and does not alter the model or its accuracy.

Figure 4: Enriched output for one recording: the v4 step timeline (colored by SAM3 category), a component inventory, and per-step SAM3 mask overlays (bright outlines). SAM3 supplies the *where*; the v4 head supplies the *which/when*.

7 Results and Discussion

Figure 5 consolidates all comparisons. Three findings recur. (1) **Backbone > head for segmentation**: the SSv2-giant features help every head, and fusion helps further. (2) **The two wins are orthogonal**: better features (fusion) and a better head (DiffAct) combine additively into v4. (3) **Correctness is data-limited**: neither larger encoders nor foundation-model VLMs move it; the lever is more error data and sharper labels. SAM3 enrichment is a pragmatic response—it makes the output visibly richer and enables non-data-limited fault signals (omission, rework), while honestly not solving the wrong-install problem.

Ego-centric Procedure Step Recognition — all architecture comparisons

IndustReal & MECCANO · ★ = best in category · metrics are % (higher better, except latency)

A Offline step segmentation
IndustReal test (32 recs) · higher is better

Configuration	Acc	Edit	F1@10	F1@25	F1@50
v1 Huge-K710 + ASFormer (baseline)	73.3	68.8	72.9	68.6	56.3
v1 + Viterbi	73.4	74.2	78.6	73.9	62.2
v2 SSv2-giant + ASFormer	73.8	71.8	73.8	71.4	59.4
v2 SSv2-giant + ASFormer + Viterbi	74.0	77.3	80.0	76.5	66.5
v2 SSv2-giant + DiffAct (prev. best)	**74.4**	**79.6**	**81.1**	**77.6**	**68.9**
Fusion (giant + IV2-B14, 2176-d) + ASFormer	75.8	73.3	74.0	70.3	63.5
Fusion + ASFormer + Viterbi	75.7	77.6	79.0	75.8	68.2
★ v4 Fusion (giant + IV2-B14) + DiffAct — NEW BEST	**74.9**	**79.5**	**83.1**	**79.1**	**70.1**

B Correctness / fault detection
IndustReal · incorrect-install recall = the "done right?" metric

Configuration (type head)	Incorrect recall	Incorrect prec.	Remove recall	Verdict
Any VideoMAE-only (Huge / giant / ViT-B) + ASFormer	0.0 %	–	~68 %	no fault detection
★ Fusion giant + InternVideo2-B14 + ASFormer	**10.1 %**	**24.4 %**	**69.4 %**	**only non-zero — best**
Fusion giant + InternVideo2-L14 + ASFormer	0.0 %	–	–	bigger encoder regressed

C Real-time / streaming
IndustReal test · causal (no future frames) · L = fixed-lag

Configuration (causal / online)	Acc	Edit	F1@10	F1@25	F1@50	Latency
v3 causal ViT-B + GRU · L = 0	50.2	54.5	54.9	48.0	30.8	0.53 s
v3 L = 8	54.3	54.5	56.9	52.3	36.4	1.60 s
★ v3 L = 16 (best streaming)	**55.6**	**55.3**	**57.1**	**53.4**	**37.4**	**2.67 s**
v3 TeSTra head · L = 16 (data-hungry)	43.5	44.5	47.2	41.4	24.8	2.67 s

D Cross-dataset generalization
MECCANO test · same fusion+ASFormer recipe, no code change

Configuration (MECCANO test)	Acc	Edit	F1@10	F1@25	F1@50	Incorrect recall
Fusion + ASFormer + Viterbi (18 cls · 11 train vids)	55.0	64.5	66.2	58.9	38.0	0.0 %

v4 = the new unified architecture: fusion FEATURES (from B) + DiffAct HEAD (from A). F1@50 70.1 is the first result above 70 (prev. DiffAct 68.9, fusion+ASFormer+Viterbi 68.2).

Correctness is DATA-limited, not architecture-limited: L14 (bigger encoder) regressed to 0 %, and MECCANO (fewer error videos) also 0 %. Lever = more + sharper error labels, not more capacity.

Figure 5: Consolidated comparison across all experiments: offline segmentation, fault detection, real-time streaming, and cross-dataset generalization. Best-in-category rows are highlighted.

8 Conclusion

We presented a controlled architecture progression for egocentric procedure step recognition in which each change—Viterbi decoding, SSv2-giant features, the DiffAct head, and two-encoder fusion—improved segmentation, ending at **v4 (Fusion + DiffAct), F1@50** = 70.1. We built a causal real-time variant, confirmed cross-dataset generalization, and characterized correctness as data-limited. Finally, we added a SAM3 enrichment layer that augments the output with masks and a component inventory at no accuracy cost, and reported that promptable foundation models localize but do not judge correctness. Future work targets the data-limited correctness lever directly: sharper (segment-close) error labels, promotion of the non-data-limited *remove* class, and SAM3-based omission/rework detection against a reference bill-of-materials.

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